

MYSQL Query Flow

What is storage engine ?

In mysql the storage engine will holds the actual data and say that it handle the major storage part .

InnoDB handle :

table data
indexes
transactions
locking
redo log
crash recovery

SQL



MySQL Server Layer



InnoDB



Memory + Disk

The journey of SELECT Statement .

SELECT name
FROM employee
WHERE id = 101;

Application



Connection



Authentication



Parser



Optimizer



Execution



InnoDB



Buffer Pool



Find row



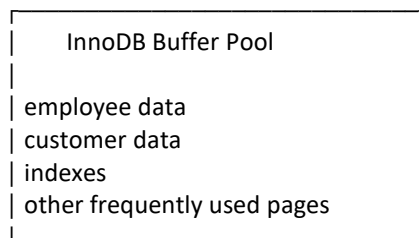
Return data



Application

Buffer Pool :

RAM



Note : Mysql frequently used data putting into the RAM to avoid the data reading from disk if we run the query to frequently .

If data found in RAM :

SELECT id=101



Buffer Pool



Found!



Return

If data not found in RAM :

SELECT



Buffer Pool



Not found



Disk se page read



Buffer Pool load



Data return

Next is INSERT :

```
INSERT INTO employee
(id, name, salary)
VALUES
(101, 'Rahul', 50000);
```

Application



MySQL



Parser



Optimizer



Execution



InnoDB



Buffer Pool

Note : Innodb modify the relevant data in memory . & not immediately put into the .ibd file .

Innodb is use the redo log file .

What is redo log ?

INSERT
↓
Memory changes
↓
Redo Log
↓
Commit

Note : Redo log basically recovery ke liye record rakhta hai ki database pages mein kya changes kiye gaye.

INSERT Rahul
↓
Server crash (note : if data are in ram then data will gone)

So :

RAM
↓
Redo Log
↓
COMMIT

Then :
Restart
↓
Redo Log
↓
Recover changes

But when data are going to .ibd file

INSERT
↓
Buffer Pool page modified
↓
Page becomes "dirty"
↓
Redo log records change
↓
Commit durability handled
↓
Later background flushing
↓
.ibd data file updated

How update query will work internally :

UPDATE employee SET salary = 60000 WHERE id = 101;

UPDATE
↓
Parser
↓
Optimizer
↓
Find row/index
↓
InnoDB
↓
Buffer Pool
↓
Modify page
↓
Redo Log
↓
Commit

And later :

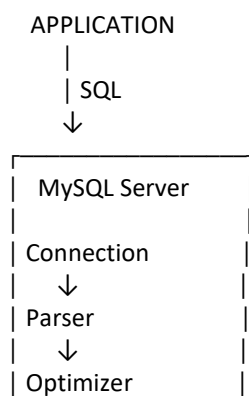
Dirty Page
↓
Disk flush
↓
.ibd

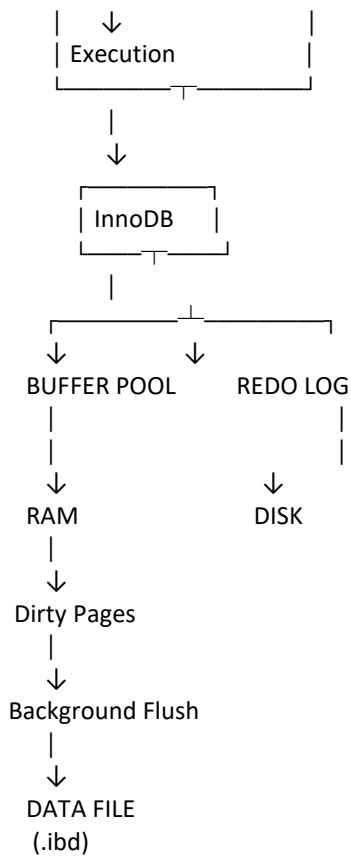
Delete :

DELETE FROM employee
WHERE id = 101;

DELETE
↓
Find row
↓
Modify InnoDB page
↓
Redo Log
↓
Commit
↓
Later disk flush

Now we have to see it daigrametically :





SELECT vs INSERT/UPDATE/DELETE

SELECT

↓
Find data
↓
Buffer Pool
↓
If missing → Disk read
↓
Return result

INSERT

↓
Create/change data
↓
Buffer Pool
↓
Redo Log
↓
Commit
↓
Later flush to data file

UPDATE

↓
Find existing data
↓
Modify
↓
Buffer Pool
↓
Redo Log
↓
Commit
↓
Later flush

DELETE

↓
Find data
↓
Modify/delete logically
↓
Buffer Pool
↓
Redo Log
↓
Commit
↓
Later flushing/purge

Simple in Hindi Sort Language .

1. Client query bhejta hai
↓
2. Connection receive karta hai
↓
3. Parser SQL samajhta hai
↓
4. Optimizer plan choose karta hai
↓
5. PRIMARY KEY index use hota hai
↓
6. Relevant B+Tree pages locate hote hain
↓
7. Page Buffer Pool mein check hota hai
↓
8. Agar RAM mein nahi → disk se read
↓
9. Page Buffer Pool mein aata hai
↓
10. Row locate hoti hai
↓
11. Result client ko return

Created By : Ajit Yadav (MSSQL , MYSQL , PostgreSQL DBA)